PART 1 · MODULE C

SQL: from SELECT to window functions

Each query is tested against a sample database.

The SQL answers below run against a small sample database of customers and orders (nine orders in three months). Each query is tested: the expected results were calculated by hand first.

C1. Total revenue by channel

SQL

```
SELECT channel, SUM(amount) AS revenue
FROM orders
GROUP BY channel
ORDER BY revenue DESC;
```

Result: search 5,600; social 3,900; email 1,600.

C2. Monthly revenue

SQL

```
SELECT substr(order_date, 1, 7) AS month, SUM(amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month;
```

Result: January 3,700; February 2,100; March 5,300.

C3. Repeat customers

SQL

```
SELECT c.name, COUNT(o.id) AS orders
FROM customers c
JOIN orders o ON o.customer_id = c.id
GROUP BY c.id, c.name
HAVING COUNT(o.id) >= 2
ORDER BY c.name;
```

Result: Asha 3 orders; Meera 2; Ravi 2.

C4. Average order value by city

SQL

```
SELECT c.city, ROUND(AVG(o.amount), 2) AS avg_order_value
FROM orders o
JOIN customers c ON c.id = o.customer_id
GROUP BY c.city
ORDER BY c.city;
```

Result: Delhi 400; Mumbai 2,000; Pune 940.

C5. Running total (window function)

SQL

```
SELECT order_date, amount,
       SUM(amount) OVER (ORDER BY order_date ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS
running_total
FROM orders
WHERE customer_id = 1
ORDER BY order_date;
```

Result: 1,200 then 2,000 then 2,700 for customer 1.

C6. Query habits

- Filter early with WHERE to reduce work.
- Index columns used in joins and filters on large tables.
- Check results against a hand calculation on a small sample.

VERIFICATION NOTE

Tool features, plans and prices change. Excel and Power BI features differ by version; check the official help pages. Links were not checked live in this draft.

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